Full-Stack Developer Intern — 48-Hour Task

Project: AI Car Mechanic Chatbot

Build a web-based chatbot where a car owner can chat with a virtual mechanic agent for troubleshooting and diagnosis.

Frontend

- React / Next.js
- Deploy on Vercel free tier
- Chat interface supporting text, image, audio and video uploads
- Show diagnosis/conversation history
- Provide a “Book Mechanic” CTA after diagnosis

Backend

- Python + Django / Django REST Framework
- Deploy on AWS free tier
- SQLite database
- REST APIs for chat, media upload, diagnosis and booking
- Use Gemini free API only where AI is actually required
- **Use traditional backend logic wherever possible and minimize AI/API usage.**

Bot Behaviour

- Act like a senior automobile technician.
- Only handle car/mechanical-related queries.
- Politely reject irrelevant queries.
- Ask relevant follow-up questions before giving a diagnosis.
- Analyze uploaded media where useful.
- After diagnosis, suggest an appropriate repair/service.
- If the customer agrees, create a mechanic booking through the backend.

Minimum APIs

- `POST /api/chat/`
- `POST /api/upload/`
- `POST /api/diagnosis/`
- `POST /api/booking/`
- `GET /api/booking/{id}/`

Deliverables

- GitHub repository
- Live frontend URL

- Live backend API URL
- README with setup instructions
- API documentation
- Short architecture explanation

Deadline: 48 hours

Evaluation: Code quality, API design, frontend UX, error handling, deployment, database design, and how effectively the candidate minimizes unnecessary AI usage.